Sama nabil

SE2

PART A Containers

A1: Run Ubuntu container

```
root@958f29b301ce: /
C:\Users\sama>mkdir lab3

C:\Users\sama> cd lab3

C:\Users\sama\lab3> docker run --rm -it --name nsdemo ubuntu:22.04 bash
Unable to find image 'ubuntu:22.04' locally
22.04: Pulling from library/ubuntu
f63eb04151bc: Pull complete
bf0d6867143e: Download complete
Digest: sha256:962f6cadeae0ea6284001009daa4cc9a8c37e75d1f5191cf0eb83fe565b63dd7
Status: Downloaded newer image for ubuntu:22.04
root@958f29b301ce:/# hostname
ps -ef
ip addr
mount | head
cat /proc/1/cgroup
958f29b301ce
UID          PID    PPID  C STIME TTY          TIME CMD
root           1       0  0 23:31 pts/0    00:00:00 bash
root          10       1  0 23:31 pts/0    00:00:00 ps -ef
bash: ip: command not found
overlay on / type overlay (rw,relatime,lowerdir=/var/lib/desktop-containerd/daemon/io.containerd.snapshotter.v1.overlayfs/snapshots/33/fs:/
ib/desktop-containerd/daemon/io.containerd.snapshotter.v1.overlayfs/snapshots/32/fs,upperdir=/var/lib/desktop-containerd/daemon/io.containe
apshotter.v1.overlayfs/snapshots/34/fs,workdir=/var/lib/desktop-containerd/daemon/io.containerd.snapshotter.v1.overlayfs/snapshots/34/work)
proc on /proc type proc (rw,nosuid,nodev,noexec,relatime)
tmpfs on /dev type tmpfs (rw,nosuid,size=65536k,mode=755)
devpts on /dev/pts type devpts (rw,nosuid,noexec,relatime,gid=5,mode=620,ptmxmode=666)
sysfs on /sys type sysfs (ro,nosuid,nodev,noexec,relatime)
cgroup on /sys/fs/cgroup type cgroup2 (ro,nosuid,nodev,noexec,relatime,nsdelegate)
mqueue on /dev/mqueue type mqueue (rw,nosuid,nodev,noexec,relatime)
shm on /dev/shm type tmpfs (rw,nosuid,nodev,noexec,relatime,size=65536k)
/dev/sde on /etc/resolv.conf type ext4 (rw,relatime)
/dev/sde on /etc/hostname type ext4 (rw,relatime)
0::/
root@958f29b301ce:/#
```

PID namespace

```
sama@DESKTOP-L3NT4QA:~$  docker ps
CONTAINER ID   IMAGE                     COMMAND                  CREATED             STATUS             PORTS                                          NA
MES
958f29b301ce   ubuntu:22.04              "bash"                   4 minutes ago       Up 4 minutes                                                      ns
demo
8b272074fb09   redis                     "docker-entrypoint.s…"   About an hour ago   Up About an hour   0.0.0.0:6380->6379/tcp, [::]:6380->6379/tcp   re
dis-node2
24f15e1b8e5e   redis                     "docker-entrypoint.s…"   About an hour ago   Up About an hour   0.0.0.0:6379->6379/tcp, [::]:6379->6379/tcp   re
dis-node1
099bde019621   quay.io/coreos/etcd:v3.5.0 "/usr/local/bin/etcd…"   About an hour ago   Up About an hour   0.0.0.0:2379->2379/tcp, [::]:2379->2379/tcp   et
cd
9396038dbdbd   ubuntu                    "/bin/bash"              2 hours ago         Up 2 hours                                                        aw
esome_euclid
33de406dce50   ubuntu                    "/bin/bash"              3 hours ago         Up 3 hours                                                        he
uristic_ishizaka
```

```
sama@DESKTOP-L3NT4QA:~$ docker inspect nsdemo --format "{{.State.Pid}}"
1505
```

```
sama@DESKTOP-L3NT4QA:~$ ps -fp 1505
UID          PID    PPID  C STIME TTY          TIME CMD
sama@DESKTOP-L3NT4QA:~$
```

cgroups limit test

```
sama@DESKTOP-L3NT4QA:~$  docker run --rm -it --cpus="0.5" --memory="256m" ubuntu:22.04 bash
root@41fe1620a54d:/#  apt-get update && apt-get install -y stress
Get:1 http://security.ubuntu.com/ubuntu jammy-security InRelease [129 kB]
Get:2 http://archive.ubuntu.com/ubuntu jammy InRelease [270 kB]
Get:3 http://security.ubuntu.com/ubuntu jammy-security/multiverse amd64 Packages [61.6 kB]
Get:4 http://security.ubuntu.com/ubuntu jammy-security/main amd64 Packages [3860 kB]
Get:5 http://archive.ubuntu.com/ubuntu jammy-updates InRelease [128 kB]
Get:6 http://archive.ubuntu.com/ubuntu jammy-backports InRelease [127 kB]
Get:7 http://archive.ubuntu.com/ubuntu jammy/main amd64 Packages [1792 kB]
Get:8 http://security.ubuntu.com/ubuntu jammy-security/universe amd64 Packages [1292 kB]
Get:9 http://archive.ubuntu.com/ubuntu jammy/universe amd64 Packages [17.5 MB]
Get:10 http://security.ubuntu.com/ubuntu jammy-security/restricted amd64 Packages [6862 kB]
Get:11 http://archive.ubuntu.com/ubuntu jammy/multiverse amd64 Packages [266 kB]
Get:12 http://archive.ubuntu.com/ubuntu jammy/restricted amd64 Packages [164 kB]
Get:13 http://archive.ubuntu.com/ubuntu jammy-updates/restricted amd64 Packages [7066 kB]
Get:14 http://archive.ubuntu.com/ubuntu jammy-updates/universe amd64 Packages [1601 kB]
Get:15 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 Packages [4188 kB]
Get:16 http://archive.ubuntu.com/ubuntu jammy-updates/multiverse amd64 Packages [69.9 kB]
Get:17 http://archive.ubuntu.com/ubuntu jammy-backports/universe amd64 Packages [35.7 kB]
Get:18 http://archive.ubuntu.com/ubuntu jammy-backports/main amd64 Packages [82.7 kB]
Fetched 45.5 MB in 15s (3128 kB/s)
Reading package lists... Done
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following NEW packages will be installed:
  stress
0 upgraded, 1 newly installed, 0 to remove and 1 not upgraded.
Need to get 18.4 kB of archives.
After this operation, 52.2 kB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu jammy/universe amd64 stress amd64 1.0.5-1 [18.4 kB]
Fetched 18.4 kB in 0s (85.4 kB/s)
debconf: delaying package configuration, since apt-utils is not installed
Selecting previously unselected package stress.
(Reading database ... 4393 files and directories currently installed.)
Preparing to unpack .../stress_1.0.5-1_amd64.deb ...
Unpacking stress (1.0.5-1) ...
Setting up stress (1.0.5-1) ...
root@41fe1620a54d:/# stress --cpu 2 --vm 1 --vm-bytes 220M --timeout 30
stress: info: [235] dispatching hogs: 2 cpu, 0 io, 1 vm, 0 hdd
stress: info: [235] successful run completed in 30s
root@41fe1620a54d:/#
```

```
sama@DESKTOP-L3NT4QA:~$  docker run --rm -it --cpus="0.5" --memory="256m" ubuntu:22.04 bash
CONTAINER ID   NAME                CPU %     MEM USAGE / LIMIT     MEM %     NET I/O         BLOCK I/O
958f29b301ce   nsdemo              0.00%     964KiB / 3.744GiB     0.02%     1.04kB / 126B   0B / 0B
8b272074fb09   redis-node2         0.27%     10.57MiB / 3.744GiB   0.28%     339kB / 499kB   6.44MB / 4.1kB
24f15e1b8e5e   redis-node1         0.29%     10.79MiB / 3.744GiB   0.28%     569kB / 385kB   2.48MB / 16.4kB
099bde019621   etcd                0.67%     9.09MiB / 3.744GiB    0.24%     872B / 126B     471kB / 0B
9396038dbdbd   awesome_euclid      0.00%     928KiB / 3.744GiB     0.02%     1.17kB / 126B   299kB / 0B
33de406dce50   heuristic_ishizaka  0.00%     972KiB / 3.744GiB     0.02%     2.12kB / 126B   81.9kB / 0B
```

PART B — Docker images

```
C:\Users\sama\lab3>docker build -f Dockerfile.basic -t lab3-basic .
[+] Building 12.7s (10/10) FINISHED                                                                    docker:desktop-linux
 => [internal] load build definition from Dockerfile.basic                                                                0.0s
 => => transferring dockerfile: 210B                                                                                      0.0s
 => [internal] load metadata for docker.io/library/python:3.12-slim                                                       0.6s
 => [internal] load .dockerignore                                                                                         0.0s
 => => transferring context: 2B                                                                                           0.0s
 => [internal] load build context                                                                                         0.0s
 => => transferring context: 84B                                                                                          0.0s
 => [1/5] FROM docker.io/library/python:3.12-slim@sha256:520153e2deb359602c9cffd84e491e3431d76e7bf95a3255c9ce9433b76ab99a    5.2s
 => => resolve docker.io/library/python:3.12-slim@sha256:520153e2deb359602c9cffd84e491e3431d76e7bf95a3255c9ce9433b76ab99a    0.0s
 => => sha256:3fc9d9ab5045181fbd49d1d30a2dbad144172d4c3c141f70117ddb0b918d8357 248B / 248B                                0.5s
 => => sha256:4d873bcef452ab1c054b9a0f25ace04a446c558a7b925aed2fc5bc21533e0f65 12.10MB / 12.10MB                          4.5s
 => => sha256:cb8c0a9140ac107575972113eb5fd2e6c3cdddc592371a5ac31c4623dcb13d5d 1.29MB / 1.29MB                            1.2s
 => => extracting sha256:cb8c0a9140ac107575972113eb5fd2e6c3cdddc592371a5ac31c4623dcb13d5d                                 0.2s
 => => extracting sha256:4d873bcef452ab1c054b9a0f25ace04a446c558a7b925aed2fc5bc21533e0f65                                 0.6s
 => => extracting sha256:3fc9d9ab5045181fbd49d1d30a2dbad144172d4c3c141f70117ddb0b918d8357                                 0.0s
 => [2/5] WORKDIR /app                                                                                                    0.1s
 => [3/5] COPY requirements.txt .                                                                                         0.0s
 => [4/5] RUN pip install -r requirements.txt                                                                             5.3s
 => [5/5] COPY app.py .                                                                                                   0.0s
 => exporting to image                                                                                                    1.2s
 => => exporting layers                                                                                                   0.7s
 => => exporting manifest sha256:865391d791e941c0467adc544048ac8e7173e26850b84e3012a1cd720a3a13c9                         0.0s
 => => exporting config sha256:6c0848d1b572a0491b7d760c3071ed18ffda2727252f4ca1a9e59d06cb5e8eab                           0.0s
 => => exporting attestation manifest sha256:bb257cba2ea60e90e614f8450759b484de9ff363924ffaf910d1aee806899a29             0.0s
 => => exporting manifest list sha256:725bc71ad7b9be1b4fab2faff047a1df1e5514a6c775dadd3da08ae763177800                    0.0s
 => => naming to docker.io/library/lab3-basic:latest                                                                      0.0s
 => => unpacking to docker.io/library/lab3-basic:latest                                                                   0.4s

C:\Users\sama\lab3>docker history lab3-basic
IMAGE          CREATED          CREATED BY                                      SIZE      COMMENT
725bc71ad7b9   23 seconds ago   CMD ["python" "app.py"]                         0B        buildkit.dockerfile.v0
<missing>      23 seconds ago   EXPOSE [5000/tcp]                               0B        buildkit.dockerfile.v0
<missing>      23 seconds ago   COPY app.py . # buildkit                        8.19kB    buildkit.dockerfile.v0
<missing>      23 seconds ago   RUN /bin/sh -c pip install -r requirements.t…   16.5MB    buildkit.dockerfile.v0
<missing>      28 seconds ago   COPY requirements.txt . # buildkit              12.3kB    buildkit.dockerfile.v0
<missing>      28 seconds ago   WORKDIR /app                                    8.19kB    buildkit.dockerfile.v0
<missing>      22 hours ago     CMD ["python3"]                                 0B        buildkit.dockerfile.v0
<missing>      22 hours ago     RUN /bin/sh -c set -eux;  for src in idle3 p…   16.4kB    buildkit.dockerfile.v0
<missing>      22 hours ago     RUN /bin/sh -c set -eux;   savedAptMark="$(a…   41.3MB    buildkit.dockerfile.v0
<missing>      22 hours ago     ENV PYTHON_SHA256=c08bc65a81971c1dd578318282…   0B        buildkit.dockerfile.v0
<missing>      22 hours ago     ENV PYTHON_VERSION=3.12.13                      0B        buildkit.dockerfile.v0
<missing>      22 hours ago     ENV GPG_KEY=7169605F62C751356D054A26A821E680…   0B        buildkit.dockerfile.v0
```

```
C:\Users\sama\lab3>docker image ls
                                                                                                  i Info →   U  In Use
IMAGE                          ID             DISK USAGE   CONTENT SIZE   EXTRA
lab3-basic:latest              725bc71ad7b9        199MB           49MB
quay.io/coreos/etcd:v3.5.0     28759af54acd        156MB         40.6MB    U
redis:latest                   d372cf7cd5ab        204MB         55.3MB    U
ubuntu:22.04                   962f6cadeae0        119MB         31.7MB    U
ubuntu:latest                  c4a8d5503dfb        119MB         31.7MB    U
```

Build basic image

```
C:\Users\sama\lab3>docker build -f Dockerfile.basic -t lab3-basic .
[+] Building 12.7s (10/10) FINISHED                                                                                  docker:desktop-linux
 => [internal] load build definition from Dockerfile.basic                                                                          0.0s
 => => transferring dockerfile: 210B                                                                                                0.0s
 => [internal] load metadata for docker.io/library/python:3.12-slim                                                                 0.6s
 => [internal] load .dockerignore                                                                                                   0.0s
 => => transferring context: 2B                                                                                                     0.0s
 => [internal] load build context                                                                                                   0.0s
 => => transferring context: 84B                                                                                                    0.0s
 => [1/5] FROM docker.io/library/python:3.12-slim@sha256:520153e2deb359602c9cffd84e491e3431d76e7bf95a3255c9ce9433b76ab99a           5.2s
 => => resolve docker.io/library/python:3.12-slim@sha256:520153e2deb359602c9cffd84e491e3431d76e7bf95a3255c9ce9433b76ab99a           0.0s
 => => sha256:3fc9d9ab5045181fbd49d1d30a2dbad144172d4c3c141f70117ddb0b918d8357 248B / 248B                                         0.5s
 => => sha256:4d873bcef452ab1c054b9a0f25ace04a446c558a7b925aed2fc5bc21533e0f65 12.10MB / 12.10MB                                   4.5s
 => => sha256:cb8c0a9140ac107575972113eb5fd2e6c3cdddc592371a5ac31c4623dcb13d5d 1.29MB / 1.29MB                                     1.2s
 => => extracting sha256:cb8c0a9140ac107575972113eb5fd2e6c3cdddc592371a5ac31c4623dcb13d5d                                           0.2s
 => => extracting sha256:4d873bcef452ab1c054b9a0f25ace04a446c558a7b925aed2fc5bc21533e0f65                                           0.6s
 => => extracting sha256:3fc9d9ab5045181fbd49d1d30a2dbad144172d4c3c141f70117ddb0b918d8357                                           0.0s
 => [2/5] WORKDIR /app                                                                                                              0.1s
 => [3/5] COPY requirements.txt .                                                                                                   0.0s
 => [4/5] RUN pip install -r requirements.txt                                                                                       5.3s
 => [5/5] COPY app.py .                                                                                                             0.0s
 => exporting to image                                                                                                              1.2s
 => => exporting layers                                                                                                             0.7s
 => => exporting manifest sha256:865391d791e941c0467adc544048ac8e7173e26850b84e3012a1cd720a3a13c9                                   0.0s
 => => exporting config sha256:6c0848d1b572a0491b7d760c3071ed18ffda2727252f4ca1a9e59d06cb5e8eab                                     0.0s
 => => exporting attestation manifest sha256:bb257cba2ea60e90e614f8450759b484de9ff363924ffaf910d1aee806899a29                       0.0s
 => => exporting manifest list sha256:725bc71ad7b9be1b4fab2faff047a1df1e5514a6c775dadd3da08ae763177800                              0.0s
 => => naming to docker.io/library/lab3-basic:latest                                                                                0.0s
 => => unpacking to docker.io/library/lab3-basic:latest                                                                             0.4s

C:\Users\sama\lab3>docker history lab3-basic
IMAGE          CREATED         CREATED BY                                      SIZE      COMMENT
725bc71ad7b9   23 seconds ago  CMD ["python" "app.py"]                         0B        buildkit.dockerfile.v0
<missing>      23 seconds ago  EXPOSE [5000/tcp]                               0B        buildkit.dockerfile.v0
<missing>      23 seconds ago  COPY app.py . # buildkit                        8.19kB    buildkit.dockerfile.v0
<missing>      23 seconds ago  RUN /bin/sh -c pip install -r requirements.t…   16.5MB    buildkit.dockerfile.v0
<missing>      28 seconds ago  COPY requirements.txt . # buildkit              12.3kB    buildkit.dockerfile.v0
<missing>      28 seconds ago  WORKDIR /app                                    8.19kB    buildkit.dockerfile.v0
<missing>      22 hours ago    CMD ["python3"]                                 0B        buildkit.dockerfile.v0
<missing>      22 hours ago    RUN /bin/sh -c set -eux;  for src in idle3 p…   16.4kB    buildkit.dockerfile.v0
<missing>      22 hours ago    RUN /bin/sh -c set -eux;   savedAptMark="$(a…   41.3MB    buildkit.dockerfile.v0
<missing>      22 hours ago    ENV PYTHON_SHA256=c08bc65a81971c1dd578318282…   0B        buildkit.dockerfile.v0
<missing>      22 hours ago    ENV PYTHON_VERSION=3.12.13                      0B        buildkit.dockerfile.v0
<missing>      22 hours ago    ENV GPG_KEY=7169605F62C751356D054A26A821E680…   0B        buildkit.dockerfile.v0
```

Build multi-stage

```
C:\Users\sama\lab3>docker build -f Dockerfile.multistage -t lab3-multi .
[+] Building 7.5s (13/13) FINISHED                                                                                                  docker:desktop-linux
 => [internal] load build definition from Dockerfile.multistage                                                                                    0.0s
 => => transferring dockerfile: 330B                                                                                                               0.0s
 => [internal] load metadata for docker.io/library/python:3.12-slim                                                                                1.0s
 => [auth] library/python:pull token for registry-1.docker.io                                                                                      0.0s
 => [internal] load .dockerignore                                                                                                                  0.0s
 => => transferring context: 2B                                                                                                                    0.0s
 => [internal] load build context                                                                                                                  0.0s
 => => transferring context: 60B                                                                                                                   0.0s
 => CACHED [builder 1/4] FROM docker.io/library/python:3.12-slim@sha256:520153e2deb359602c9cffd84e491e3431d76e7bf95a3255c9ce9433b76ab99a           0.0s
 => => resolve docker.io/library/python:3.12-slim@sha256:520153e2deb359602c9cffd84e491e3431d76e7bf95a3255c9ce9433b76ab99a                          0.0s
 => CACHED [stage-1 2/4] WORKDIR /app                                                                                                              0.0s
 => [builder 2/4] WORKDIR /build                                                                                                                   0.0s
 => [builder 3/4] COPY requirements.txt .                                                                                                          0.0s
 => [builder 4/4] RUN pip install --prefix=/install -r requirements.txt                                                                            5.2s
 => [stage-1 3/4] COPY --from=builder /install /usr/local                                                                                          0.1s
 => [stage-1 4/4] COPY app.py .                                                                                                                    0.1s
 => exporting to image                                                                                                                             0.5s
 => => exporting layers                                                                                                                            0.3s
 => => exporting manifest sha256:48fa8ce81d85d60c58b770cc718ab0eb80086cb68b9d6a1a293f278c8b1a1ffa                                                  0.0s
 => => exporting config sha256:1476dd372cb5cda0234ef0de0b09a6ac17c4b24667f2ab0bd7d3d3d85fd4df86                                                    0.0s
 => => exporting attestation manifest sha256:caeb26f5c78aed35ef39a5bf730817371be844127e79d20403cff73ca520a069                                      0.0s
 => => exporting manifest list sha256:255417445d898060b940f4882cf32ef2cae2e22b30d1a7167ab80fb6f9e389ea                                             0.0s
 => => naming to docker.io/library/lab3-multi:latest                                                                                               0.0s
 => => unpacking to docker.io/library/lab3-multi:latest                                                                                            0.1s

C:\Users\sama\lab3>docker history lab3-multi
IMAGE          CREATED        CREATED BY                                      SIZE      COMMENT
255417445d89   2 seconds ago  CMD ["python" "app.py"]                          0B        buildkit.dockerfile.v0
<missing>      2 seconds ago  EXPOSE [5000/tcp]                               0B        buildkit.dockerfile.v0
<missing>      2 seconds ago  COPY app.py . # buildkit                        8.19kB    buildkit.dockerfile.v0
<missing>      3 seconds ago  COPY /install /usr/local # buildkit             5.51MB    buildkit.dockerfile.v0
<missing>      4 minutes ago  WORKDIR /app                                    8.19kB    buildkit.dockerfile.v0
<missing>      22 hours ago   CMD ["python3"]                                 0B        buildkit.dockerfile.v0
<missing>      22 hours ago   RUN /bin/sh -c set -eux;  for src in idle3 p…   16.4kB    buildkit.dockerfile.v0
<missing>      22 hours ago   RUN /bin/sh -c set -eux;   savedAptMark="$(a…   41.3MB    buildkit.dockerfile.v0
<missing>      22 hours ago   ENV PYTHON_SHA256=c08bc65a81971c1dd578318282…   0B        buildkit.dockerfile.v0
<missing>      22 hours ago   ENV PYTHON_VERSION=3.12.13                      0B        buildkit.dockerfile.v0
<missing>      22 hours ago   ENV GPG_KEY=7169605F62C751356D054A26A821E680…   0B        buildkit.dockerfile.v0
<missing>      22 hours ago   RUN /bin/sh -c set -eux;  apt-get update;  a…   4.94MB    buildkit.dockerfile.v0
<missing>      22 hours ago   ENV LANG=C.UTF-8                                0B        buildkit.dockerfile.v0
<missing>      22 hours ago   ENV PATH=/usr/local/bin:/usr/local/sbin:/usr…   0B        buildkit.dockerfile.v0
<missing>      2 days ago     # debian.sh --arch 'amd64' out/ 'trixie' '@1…   87.4MB    debuerreotype 0.17

C:\Users\sama\lab3>docker image ls
                                                                                                                                  i Info →   U  In Use
IMAGE                       ID             DISK USAGE   CONTENT SIZE   EXTRA
lab3-basic:latest           725bc71ad7b9       199MB           49MB
lab3-multi:latest           255417445d89       184MB         44.7MB
quay.io/coreos/etcd:v3.5.0  28759af54acd       156MB         40.6MB    U
redis:latest                d372cf7cd5ab       204MB         55.3MB    U
ubuntu:22.04                962f6cadeae0       119MB         31.7MB    U
ubuntu:latest               c4a8d5503dfb       119MB         31.7MB    U

C:\Users\sama\lab3>
```

Which is smaller?

multi-stage is smaller

Why?

- builder stage is not included in final image

Why layer order matters?

- Docker caches layers
- changing app.py only rebuilds last layer
-

PART C Kubernetes (kind)

PODS LIST

```
C:\Users\sama\lab3>kubectl get pods
NAME                          READY   STATUS    RESTARTS   AGE
lab3-web-67c8dcdc99-6pqgz     1/1     Running   0          5s
lab3-web-67c8dcdc99-bgsp8     1/1     Running   0          5s
lab3-web-67c8dcdc99-ldzhl     1/1     Running   0          5s
```

This shows that Kubernetes Deployment created 3 replicas of the application. Each Pod is independently running and managed by the scheduler.

DEPLOYMENT

```
C:\Users\sama>kubectl get deployments
NAME        READY   UP-TO-DATE   AVAILABLE   AGE
lab3-web    3/3     3            3           6m37s
```

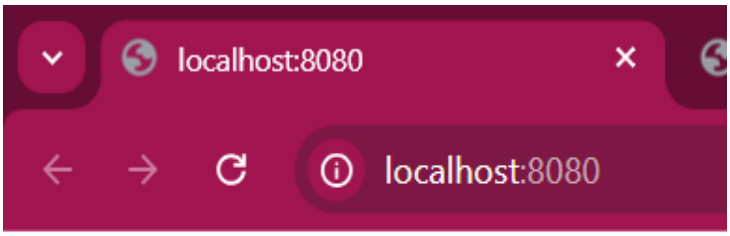
```
C:\Users\sama> kubectl describe deployment lab3-web
Name:                   lab3-web
Namespace:              default
CreationTimestamp:      Thu, 23 Apr 2026 02:41:31 +0200
Labels:                 <none>
Annotations:            deployment.kubernetes.io/revision: 1
Selector:               app=lab3-web
Replicas:               3 desired | 3 updated | 3 total | 3 available | 0 unavailable
StrategyType:           RollingUpdate
MinReadySeconds:        0
RollingUpdateStrategy:  25% max unavailable, 25% max surge
Pod Template:
  Labels:  app=lab3-web
  Containers:
   web:
    Image:         lab3-multi
    Port:          5000/TCP
    Host Port:     0/TCP
    Environment:   <none>
    Mounts:        <none>
  Volumes:         <none>
  Node-Selectors:  <none>
  Tolerations:     <none>
Conditions:
  Type           Status  Reason
  ----           ------  ------
  Available      True    MinimumReplicasAvailable
  Progressing    True    NewReplicaSetAvailable
OldReplicaSets:  <none>
NewReplicaSet:   lab3-web-67c8dcdc99 (3/3 replicas created)
Events:
  Type    Reason             Age    From                   Message
  ----    ------             ----   ----                   -------
  Normal  ScalingReplicaSet  6m44s  deployment-controller  Scaled up replica set lab3-web-67c8dcdc99 f
```

The Deployment defines the desired state (3 replicas). Kubernetes automatically ensures the correct number of Pods are running and replaces any failed ones.

BROWSER / CURL OUTPUT

```
C:\Users\sama\lab3>kubectl port-forward service/lab3-web-svc 8080:80
Forwarding from 127.0.0.1:8080 -> 5000
Forwarding from [::1]:8080 -> 5000
Handling connection for 8080
Handling connection for 8080
E0423 02:40:30.461340   18228 portforward.go:424] "Unhandled Error" err="an error occurred forwarding 80
o pod b84e12979fbe1becbea9183dbef4b9c4bc0608ba6c5e2a6a44ee549854dec4ef, uid : failed to execute portforw
s/cni-44cc875c-4a85-587e-e638-cc4911ce78c2\": failed to connect to localhost:5000 inside namespace \"b84
e2a6a44ee549854dec4ef\", IPv4: dial tcp4 127.0.0.1:5000: connect: connection refused IPv6 dial tcp6 [::1
E0423 02:40:30.461891   18228 portforward.go:424] "Unhandled Error" err="an error occurred forwarding 80
o pod b84e12979fbe1becbea9183dbef4b9c4bc0608ba6c5e2a6a44ee549854dec4ef, uid : failed to execute portforw
s/cni-44cc875c-4a85-587e-e638-cc4911ce78c2\": failed to connect to localhost:5000 inside namespace \"b84
e2a6a44ee549854dec4ef\", IPv4: dial tcp4 127.0.0.1:5000: connect: connection refused IPv6 dial tcp6 [::1
```

ok



Hello from Lab 3

```
C:\Users\sama>curl http://localhost:8080/
Hello from Lab 3
C:\Users\sama>curl http://localhost:8080/health
ok
```

PART D Scheduling

kubectl get nodes --show-labels

```
C:\Users\sama\lab3>kubectl get nodes --show-labels
NAME                        STATUS   ROLES           AGE   VERSION   LABELS
lab3-cluster-control-plane   Ready    control-plane   30m   v1.35.0   beta.kubernetes.io/arch=amd64,beta.kubernetes.io/o
s=linux,kubernetes.io/arch=amd64,kubernetes.io/hostname=lab3-cluster-control-plane,kubernetes.io/os=linux,node-role.kube
rnetes.io/control-plane=,node-role=general
```

updated pods

```
C:\Users\sama\lab3>kubectl get pods -o wide
NAME                         READY   STATUS    RESTARTS   AGE     IP           NODE                        NOMINATED NO
DE   READINESS GATES
lab3-web-7bb75499ff-ghqhj   1/1     Running   0          3m18s   10.244.0.12   lab3-cluster-control-plane   <none>
     <none>
lab3-web-7bb75499ff-jvz96   1/1     Running   0          3m20s   10.244.0.11   lab3-cluster-control-plane   <none>
     <none>
lab3-web-7bb75499ff-kbj7w   1/1     Running   0          3m17s   10.244.0.13   lab3-cluster-control-plane   <none>
     <none>
```

PART E — Self-healing

Delete pod

```
Microsoft Windows [Version 10.0.26200.8246]
(c) Microsoft Corporation. All rights reserved.

C:\Users\sama> cd C:\Users\sama\lab3

C:\Users\sama\lab3>kubectl delete pod lab3-web-7bb75499ff-67fmm
pod "lab3-web-7bb75499ff-67fmm" deleted from default namespace

C:\Users\sama\lab3>kubectl get pods -w
NAME                         READY   STATUS    RESTARTS   AGE
lab3-web-7bb75499ff-6mwl7   1/1     Running   0          31s
lab3-web-7bb75499ff-jvz96   1/1     Running   0          9m20s
lab3-web-7bb75499ff-kbj7w   1/1     Running   0          9m17s
```

```
C:\Users\sama\lab3>kubectl get pods -w
NAME                         READY   STATUS    RESTARTS   AGE
lab3-web-7bb75499ff-67fmm   1/1     Running   0          107s
lab3-web-7bb75499ff-jvz96   1/1     Running   0          8m12s
lab3-web-7bb75499ff-kbj7w   1/1     Running   0          8m9s
lab3-web-7bb75499ff-67fmm   1/1     Terminating   0          2m24s
lab3-web-7bb75499ff-67fmm   1/1     Terminating   0          2m24s
lab3-web-7bb75499ff-6mwl7   0/1     Pending       0          0s
lab3-web-7bb75499ff-6mwl7   0/1     Pending       0          0s
lab3-web-7bb75499ff-6mwl7   0/1     ContainerCreating   0          0s
lab3-web-7bb75499ff-6mwl7   1/1     Running             0          1s
```

Probes

```
C:\Users\sama\lab3>kubectl apply -f probe-deployment.yaml
deployment.apps/lab3-web-probe created
```

```
C:\Users\sama\lab3>kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
lab3-web-7bb75499ff-6mwl7           1/1     Running   0          9m15s
lab3-web-7bb75499ff-jvz96           1/1     Running   0          18m
lab3-web-7bb75499ff-kbj7w           1/1     Running   0          18m
lab3-web-probe-6f45cc9c66-q2z55     1/1     Running   0          4m37s
lab3-web-probe-6f45cc9c66-rd9m6     1/1     Running   0          4m37s
```

readiness + liveness probe section

```
C:\Users\sama\lab3>kubectl describe pod lab3-web-probe-6f45cc9c66-q2z55
Name:             lab3-web-probe-6f45cc9c66-q2z55
Namespace:        default
Priority:         0
Service Account:  default
Node:             lab3-cluster-control-plane/172.19.0.2
Start Time:       Thu, 23 Apr 2026 03:17:02 +0200
Labels:           app=lab3-web-probe
                  pod-template-hash=6f45cc9c66
Annotations:      <none>
Status:           Running
IP:               10.244.0.17
IPs:
  IP:           10.244.0.17
Controlled By:  ReplicaSet/lab3-web-probe-6f45cc9c66
Containers:
  web:
    Container ID:   containerd://57d26122594fe8d5c590b0d6703741629406235c0cd3de566792a8499d781309
    Image:          lab3-multi
    Image ID:       sha256:8608f86843be6c7e8ecc864c4150707c5522baf3c2f5cb0a56d4c8e9a1debb84
    Port:           5000/TCP
    Host Port:      0/TCP
    State:          Running
      Started:      Thu, 23 Apr 2026 03:17:03 +0200
    Ready:          True
    Restart Count:  0
    Liveness:       http-get http://:5000/health delay=10s timeout=1s period=10s #success=1 #failure=3
    Readiness:      http-get http://:5000/health delay=3s timeout=1s period=5s #success=1 #failure=3
    Environment:    <none>
    Mounts:
      /var/run/secrets/kubernetes.io/serviceaccount from kube-api-access-gt52t (ro)
Conditions:
  Type                        Status
  PodReadyToStartContainers   True
  Initialized                 True
  Ready                       True
  ContainersReady             True
  PodScheduled                True
Volumes:
  kube-api-access-gt52t:
    Type:                    Projected (a volume that contains injected data from multiple sources)
    TokenExpirationSeconds:  3607
    ConfigMapName:           kube-root-ca.crt
```

Kill process inside container using container ID with watch

```
C:\Users\sama\lab3>kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
lab3-web-7bb75499ff-6mwl7           1/1     Running   0          17m
lab3-web-7bb75499ff-jvz96           1/1     Running   0          26m
lab3-web-7bb75499ff-kbj7w           1/1     Running   0          26m
lab3-web-probe-6f45cc9c66-q2z55     1/1     Running   0          12m
lab3-web-probe-6f45cc9c66-rd9m6     1/1     Running   0          12m

C:\Users\sama\lab3>kubectl exec -it lab3-web-7bb75499ff-kbj7w    -- /bin/sh
# kill 1
# exit
```

```
C:\Users\sama> cd C:\Users\sama\lab3

C:\Users\sama\lab3> docker exec -it lab3-cluster-control-plane crictl stop 57d26122594fe8d5c590b0d6703741629406235c0cd3d
e566792a8499d781309
57d26122594fe8d5c590b0d6703741629406235c0cd3de566792a8499d781309
```

```
C:\Users\sama\lab3> kubectl get pods -w
NAME                                READY   STATUS    RESTARTS   AGE
lab3-web-7bb75499ff-6mwl7           1/1     Running   0          27m
lab3-web-7bb75499ff-jvz96           1/1     Running   0          36m
lab3-web-7bb75499ff-kbj7w           1/1     Running   0          36m
lab3-web-probe-6f45cc9c66-q2z55     1/1     Running   0          23m
lab3-web-probe-6f45cc9c66-rd9m6     1/1     Running   0          23m
lab3-web-probe-6f45cc9c66-q2z55     0/1     Error     0          24m
lab3-web-probe-6f45cc9c66-q2z55     0/1     Running   1 (2s ago)   24m
lab3-web-probe-6f45cc9c66-q2z55     1/1     Running   1 (8s ago)   24m
```

To test Kubernetes self-healing behavior, a running container inside a Pod was manually stopped using the container runtime inside the kind cluster node (crictl stop <container-id>). After the container was terminated, Kubernetes immediately detected the failure and recreated a replacement container to maintain the desired number of replicas defined in the Deployment.

Optional extension for advanced students

```
C:\Users\sama\lab3> kubectl get nodes
NAME                        STATUS   ROLES           AGE     VERSION
lab3-multinode-control-plane   Ready    control-plane   9m32s   v1.35.0
lab3-multinode-worker          Ready    <none>          9m20s   v1.35.0
lab3-multinode-worker2         Ready    <none>          9m20s   v1.35.0

C:\Users\sama\lab3>kubectl get pods -o wide
NAME                       READY   STATUS    RESTARTS   AGE     IP       NODE     NOMINATED NODE   READINESS GATES
lab3-web-549c6c6b7f-kgf8x   0/1     Pending   0          5m39s   <none>   <none>   <none>           <none>
lab3-web-549c6c6b7f-llqkc   0/1     Pending   0          5m39s   <none>   <none>   <none>           <none>
lab3-web-549c6c6b7f-s79wz   0/1     Pending   0          5m39s   <none>   <none>   <none>           <none>
```

```
C:\Users\sama\lab3>kubectl taint nodes lab3-multinode-worker2 key=restricted:NoSchedule
node/lab3-multinode-worker2 tainted
```

Questions:

1. Why do namespaces alone not guarantee fair resource use?

Namespaces only separate what each container can *see*, not what it is *allowed to use*.
So even if containers look isolated, one container can still take all the CPU or memory if nothing is limiting it.

---

2. How do cgroups improve cluster stability?

cgroups make sure each container gets a fair share of system resources.
They stop one container from using too much CPU or memory, which helps prevent the whole system from slowing down or crashing.

---

3. Why is Docker image layering important for large-scale orchestration?

Layering makes Docker images more efficient. Shared layers can be reused, so you don't rebuild or store everything from scratch every time.
This saves time, space, and makes deploying many containers much faster.

---

4. What does Kubernetes mean by desired state?

Desired state is basically what you *want* your system to look like, written in YAML.
Kubernetes constantly checks if reality matches that and automatically fixes things if they don't.

---

5. How is self-healing different from traditional manual operations?

In traditional systems, a person has to notice a failure and fix it manually.
In Kubernetes, the system automatically detects problems and replaces or restarts containers without waiting for human intervention.

---

6. Why are readiness probes and liveness probes not interchangeable?

They check different things.
Readiness probes decide if a pod is ready to receive traffic, while liveness probes check if it is still working properly.
One affects traffic routing, the other can restart the container.

---

7. What is one limitation of observing scheduling in a single-node kind cluster compared with a real multi-node cluster?

In a single-node kind cluster, everything runs on one machine, so you don't really see real scheduling decisions.
In a real multi-node cluster, Kubernetes has to choose between multiple machines, which affects load balancing and placement — something you can't properly observe in kind.